



UNIVERSITY OF CAPE TOWN
Department of Electrical Engineering
EEE4022F/S - Final Year Project
Graduate Attribute Tracking Form

Student name:	Samson Okuthe	DP Awarded? [Y/N]	
Student no:	OKTSAM001	Supervisor name:	Simon Winberg
Date:	September 23, 2026	Date:	September 23, 2026
Student signature:		Supervisor signature:	

Instructions:

Students must explain in this document what they **have already done** and what they **plan to do** to satisfy each Graduate Attribute. Descriptions of each GA is provided at the end of the document. Supervisors respond to the student's plans and current progress, providing additional comments or advice as they see fit. Once the student's progress is deemed sufficient (a few weeks before submission at the due date for this form), supervisors indicate that DP can be awarded.

VERY IMPORTANT: Receiving DP for the course does NOT imply that all GA's have been met in the course. Assessment of GA's only happen in the final marking of the project report.

GA 1: Problem Solving

Student Response:

Formulated the core research question around measurable vehicle dynamic response rather than subjective ride comfort, rejecting a driver-comfort framing for one suited to signal-processing analysis. Root-caused several faults from first principles: a GPIO conflict resolved via the manufacturer's schematic; a BLE service-discovery failure confirmed against an independent reference implementation; a systematic measurement flaw caught before any field data existed to be corrupted by it, where literature review showed a raw acceleration RMS conflates road roughness with vehicle speed, since the road-to-acceleration transfer function is itself speed dependent. Conducted a competitive-landscape and literature audit to formulate a specific, evidenced research gap: synchronised multi-sensor vehicle-dynamics characterisation for a three-wheeled cargo vehicle, distinct from existing four-wheel/motorcycle telemetry work and existing commercial telematics-plus-HUD products. The next stage is to run this characterisation on the moving vehicle rather than the bench, since every subsystem validated so far has been proven individually but not yet on the vehicle the research question is actually about.

Supervisor Response:

GA 4: Investigations, Experiments, and Data Analysis

Student Response:

Conducted hypothesis-test-conclusion investigations throughout: the GNSS field-count assumption (hypothesised 13 fields, tested live, found 9, corrected); the BMS service-UUID assumption (tested against a powered pack, found wrong, corrected, cross-validated against an independent parser); the sync-pulse timing offset (bounded to under one second via a lens-cover test). Designed a controlled experiment for the ride-characterisation feature (a synthetic sine wave with analytically-known RMS/peak-to-peak/crest-factor values) validating the computation independently of sensor hardware. A further investigation found two nominally-identical accelerometers disagreeing by 7% at rest; hypothesised as factory tolerance, corrected via a per-sensor calibration, confirmed to hold under an induced dynamic disturbance too, not only the static case it was derived from. A continuously-run literature pipeline has retrieved 733 candidate sources and independently verified 120 of them. The next stage is to bench-validate the newly sourced VL53L5CX proximity sensor against an independent published characterisation of the same chip before trusting its datasheet figures, and to capture a genuine outdoor GNSS fix to resolve the one remaining untested field-mapping assumption.

Supervisor Response:

GA 5: Use of Engineering Tools

Student Response:

Applied and recognised limitations of: PlatformIO/Arduino/FreeRTOS for embedded development; `esptool` and serial diagnostics for hardware verification; OpenSCAD for a parametric, printable camera-mount design; and Python automation (`paramiko`, `PyMuPDF` for schematic verification, `PIL` for frame analysis) built to support testing. Tool limitations were explicitly verified rather than assumed: not claiming a BMS checksum formula correct merely because a library accepts it, with no vendor datasheet to confirm it independently; and an OpenSCAD export metric initially misread as a mount-design fault, re-checked against a known-valid control case (a plain reference cube) before accepting or rejecting the concern. The next stage is to complete LittleFS replay-on-reconnect logic and integrate the VL53L5CX proximity sensor into the existing per-task acquisition architecture once it arrives.

Supervisor Response:

GA 6: Professional and Technical Communication

Student Response:

Maintained a continuously-updated Methodology chapter documenting engineering work honestly, including faults found and their resolution. Communicated professionally in writing with the Controls Lab custodian to negotiate shared vehicle access. Identified and corrected a cross-chapter inconsistency before it reached a reader: the Introduction described the camera's object-detection component as an active in-scope feature, while the Methodology chapter had already documented it as implemented but withheld under the project's ethics constraint; revised the Introduction to match rather than leaving the contradiction for a marker to find. The next stage is to draft the Results chapter directly from the evidence already logged in Methodology, maintaining the same documentation discipline through to submission.

Supervisor Response:

GA 8: Individual Working

Student Response:

Took full individual responsibility for hardware bring-up, firmware debugging, and report authorship. Respected shared-resource constraints: avoided modifying the inherited dashboard's SD card and an existing vehicle-access schedule reserved for another team. The next stage is to schedule and complete the outstanding battery test and the first on-vehicle field run, both currently awaiting lab and supervisor coordination ahead of the tracking deadline.

Supervisor Response:

GA 9: Independent Learning Ability

Student Response:

Repeatedly corrected my own assumptions on independently-gathered evidence: verified GPIO pin assignments against a primary-source schematic after a placeholder proved wrong; corrected a vehicle-identity error after independently checking documentation. Built a continuously-running literature pipeline specifically to keep self-correcting rather than trust one search: it has since surfaced real prior art that forced two research-gap claims to narrow rather than stand unexamined. Separately, audited my own use of machine learning against the approved ethics application before it became a problem, and recorded the position explicitly - direct engagement with the social and ethical implications this outcome requires. The next stage is to apply the same evidence-before-assumption standard to the newly adopted proximity sensor, treating its independently published characterisation as the basis for expected performance rather than its datasheet figures alone.

Supervisor Response:

GA 1: Problem Solving

Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.

- A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

- Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.

GA 4: Investigations, Experiments and Data Analysis

Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

The balance of investigation and experiment should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

Note: An investigation differs from a design in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

GA 5: Use of engineering tools

Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.

- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

A range of techniques, resources and modern engineering and IT tools appropriate to the disciplinary designation of the programme.

GA 6: Professional and Technical Communication

Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large, taking into account cultural, language, and learning differences.

This course evaluates the long report component of this outcome at exit level. Material to be communicated is in an academic or simulated professional context. Audiences range from engineering peers, management and lay persons, using appropriate academic or professional discourse. Written reports (10 000 to 15 000 words plus tables, diagrams and appendices) should cover material at exit-level. Methods of providing information include the conventional methods of the discipline, for example engineering drawings, as well as subject-specific methods.

GA 8: Individual, Team and Multidisciplinary Working

Demonstrate competence to work effectively as an individual, in teams and in multidisciplinary environments. This course evaluates the **individual** working component of this learning outcome at exit level.

Knowledge of professional ethics, responsibilities and norms of engineering practice.

GA 9: Independent Learning Ability

Demonstrate competence to engage in independent learning through well developed learning skills.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

Operate independently in complex, ill-defined contexts requiring personal responsibility and initiative, accurately self-evaluate and take responsibility for learning requirements; be aware of social and ethical implications of applying knowledge in particular contexts.

- Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective selfevaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders.
- Reflection of self-learning to begin to recognise if what has been covered meets the needs of the activity or task.